



CSE-476: Introduction to Natural Language Processing

Quiz 1

1 Multiple Choice Questions (Single correct)

Instructions: Instructions: Select the correct option with a short explanation.

1. In the BIO tagging scheme for NER, what does the “I-” prefix indicate?
A. Beginning of an entity **B.** Inside an entity **C.** Outside any entity **D.** Between two tokens
2. Which statement best describes an HMM emission model in sequence tagging?
A. It uses only words (observations) and ignores labels **B.** It is identical to an HMM **C.** It models transitions between labels but does not consider the words **D.** It models emissions $P(\text{word} \mid \text{label})$
3. In a Hidden Markov Model (HMM) for NER, a transition probability is:
A. $P(\text{label} \mid \text{previous label})$ **B.** $P(\text{label} \mid \text{word})$ **C.** $P(\text{word} \mid \text{previous word})$ **D.** $P(\text{word} \mid \text{label})$
4. Which of the following best describes a **dependency parser**?:
A. A task that identifies predicates in a sentence and labels roles of elements such as agent, theme, or location **B.** A process that assigns grammatical roles like subject and object based on syntactic structure **C.** A method that classifies named entities and links them to a knowledge base **D.** A technique that labels words with their parts of speech and morphological features
5. Which of the following best describes **run-length encoding**? **A.** A technique that compresses text by replacing consecutive repeated characters with a count and a symbol **B.** A tokenization method that iteratively

merges the most frequent pair of symbols in a corpus **C.** A tokenization method that splits text based on whitespace and punctuation **D.** A character-level encoding scheme that assigns fixed-length binary codes to characters

6. Which of the following best describes the main goal of part-of-speech tagging in NLP?
A. To label each token with a grammatical category **B.** To convert text into a sequence of symbols suitable for modeling **C.** To reduce vocabulary size to zero **D.** To assign probabilities to sentences
7. Why is using a tokenizer designed only for English spacing conventions considered an insufficient tokenizer? **A.** It produces too many tokens **B.** It does not work well for languages without explicit word boundaries **C.** It cannot tokenize English at all **D.** It ignores punctuation entirely
8. Which of the following is NOT listed as a typical part-of-speech tagging label set?
A. Part-of-speech tagging **B.** Named Entity Recognition **C.** Dependency parsing **D.** Sentiment analysis
9. What is a misconception about BIO tagging for Named Entity Recognition?
A. It cannot represent overlapping or nested entities **B.** It requires too many labels **C.** It does not support multiple entity types **D.** It ignores entity boundaries
10. When choosing between fast but less accurate models and slow but more accurate models, which trade-off must be considered? **A.** Speed vs accuracy **B.** Tokenized sequence length vs vocabulary size **C.** Precision vs recall **D.** Memory vs computation